

COMPLEX FOURIER SERIES (PART 4)

FROM REAL TO COMPLEX FOURIER FORMULAE

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1 Introduction - Euler's Formula

Recall Euler's formula for complex numbers

$$e^{i\alpha} = \cos \alpha + i \sin \alpha, \quad (1)$$

and for $-i\alpha$ it reads

$$e^{-i\alpha} = \cos \alpha - i \sin \alpha, \quad (2)$$

and by adding (and subtracting) those equations side by side we obtain

$$\begin{aligned} \cos \alpha &= \frac{e^{i\alpha} + e^{-i\alpha}}{2}, \\ \sin \alpha &= \frac{e^{i\alpha} - e^{-i\alpha}}{2i}. \end{aligned}$$

If we put $\alpha = \frac{n\pi x}{l}$ we obtain

$$\cos \frac{n\pi x}{l} = \frac{e^{i\frac{n\pi x}{l}} + e^{-i\frac{n\pi x}{l}}}{2} \quad (3)$$

$$\sin \frac{n\pi x}{l} = \frac{e^{i\frac{n\pi x}{l}} - e^{-i\frac{n\pi x}{l}}}{2i} \quad (4)$$

2 Real Fourier Series for a $2l$ -periodic Function

Now we can implement those formulae to the real Fourier series for a $2l$ - periodic function f , which satisfies the Dirichlet conditions on $\langle -l, l \rangle$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{+\infty} a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l},$$

where as the consequence of the Euler-Fourier theorem the coefficients are given by the formulae

$$\begin{aligned} a_0 &= \frac{1}{l} \int_{-l}^l f(x) dx; \\ a_n &= \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx, \quad n \geq 1; \\ b_n &= \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx, \quad n \geq 1. \end{aligned}$$

Once (3) and (4) are implemented to formula for a_n and b_n we obtain

$$\begin{aligned} &a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} = \\ &= a_n \left(\frac{e^{i\frac{n\pi x}{l}} + e^{-i\frac{n\pi x}{l}}}{2} \right) + b_n \left(\frac{e^{i\frac{n\pi x}{l}} - e^{-i\frac{n\pi x}{l}}}{2i} \right) = \\ &= \frac{a_n - ib_n}{2} e^{i\frac{n\pi x}{l}} + \frac{a_n + ib_n}{2} e^{-i\frac{n\pi x}{l}}. \end{aligned}$$

If we introduce the following notation

$$c_0 = \frac{a_0}{2}, c_n = \frac{a_n - ib_n}{2}, c_{-n} = \frac{a_n + ib_n}{2} \quad (5)$$

we can rewrite our Fourier series as

$$c_0 + \sum_{n=1}^{+\infty} (c_n e^{i \frac{n\pi x}{l}} + c_{-n} e^{-i \frac{n\pi x}{l}}) \quad (6)$$

or briefly

$$\sum_{n=-\infty}^{+\infty} c_n e^{i \frac{n\pi x}{l}}. \quad (7)$$

Formula (7) is called complex form of a trigonometric Fourier series, or just complex Fourier series.

If the function satisfies at the interval $\langle -l, l \rangle$ Dirichlets conditions, then it can be expanded into complex Fourier series. However, to understand it thoroughly one would need to look deeper into this formula, to specify what it means that doublesided series with complex expression underneath the sum is convergent (and in what sens convergent?).

Finally we obtain

$$f(x) = \sum_{n=-\infty}^{+\infty} c_n e^{i \frac{n\pi x}{l}},$$

with coefficients defined as

$$c_0 = \frac{1}{2} \int_{-l}^l f(x) dx$$

$$c_n = \frac{1}{2l} \int_{-l}^l f(x) \left(\cos \frac{n\pi x}{l} - i \sin \frac{n\pi x}{l} \right) dx = \frac{1}{2l} \int_{-l}^l f(x) e^{-i \frac{n\pi x}{l}} dx, \text{ for } n > 0. \quad (8)$$

$$c_{-n} = \frac{1}{2l} \int_{-l}^l f(x) \left(\cos \frac{n\pi x}{l} + i \sin \frac{n\pi x}{l} \right) dx = \frac{1}{2l} \int_{-l}^l f(x) e^{i \frac{n\pi x}{l}} dx, \text{ for } n > 0. \quad (9)$$

or briefly (8) and (9) can be written in one formula as

$$c_n = \frac{1}{2l} \int_{-l}^l f(x) e^{-i \frac{n\pi x}{l}} dx, \text{ for } n = \pm 1, \pm 2, \pm 3, \dots$$

3 Complex Fourier Series for a $2l$ -periodic function

In the previous section we have derived formula for for a $2l$ - periodic function f , which satisfies the Dirichlet conditions on $\langle -l, l \rangle$

$$f(x) = \sum_{n=-\infty}^{+\infty} c_n e^{i \frac{n\pi x}{l}},$$

with coefficients defined as

$$c_n = \frac{1}{2l} \int_{-l}^l f(x) e^{-i \frac{n\pi x}{l}} dx, \text{ for } n = 0, \pm 1, \pm 2, \pm 3, \dots$$